Mechanistic Validity: Scientific Formalization and Theory of Validity for Interpretability Claims

Abstract

A mechanistic explanation of a neural network is not a single experimental result, but a structured claim linking a theoretical construct, a measurement procedure, an intervention, and a scope of interpretation. We introduce Mechanistic Validity (MechVal), a formalized epistemology for validating these causal claims, drawing on validation methodology from causal inference, measurement theory, and the philosophy of science. The framework makes interpretability claims explicit, falsifiable, and comparable, formalizing the discovery, validation, and interpretation of causal mechanisms as separate evidential stages. A central problem the framework addresses is that post-hoc mechanistic narratives are unfalsifiable: given any set of ablation results, a plausible story can always be constructed to fit them. MechVal organizes 27 operational criteria across five validity types—construct, measurement, internal, external, and interpretive—in dependency order, and assigns each claim to one of five verdict tiers from Proposed to Validated. The criteria make the absence of falsification tests visible in the validity profile. The framework is method-agnostic, applying the same criteria to circuits, SAE features, probing classifiers, steering vectors, transcoders, and crosscoders, and is supported by an open-source implementation with 84 metrics across five evidence families and 14 calibrations. Claims are additionally qualified by one of Marr's (1982) three canonical levels of description—computational, algorithmic, and implementational—which we extend with representational as a fourth level and with four implementational sub-types: topographic, connectomic, activation-statistical, and functional. Applying the framework to 13 published circuits, we show that the tier system discriminates among claims—from Proposed (knowledge neurons, probing classifiers) to Triangulated (induction heads)—with every verdict traceable to specific present or missing evidence. By making the evidential status of a claim explicit and the path to strengthening it concrete, MechVal aims to help interpretability mature into a discipline whose claims are both interpretable and scientifically validated.

1 Introduction

Mechanistic interpretability has produced a remarkable catalog of internal mechanisms: induction heads that implement in-context copying (Olsson et al., 2022), a multi-role circuit for indirect object identification (Wang et al., 2023), a greater-than algorithm (Hanna et al., 2023), and sparse autoencoder features proposed to correspond to human-interpretable concepts (Cunningham et al., 2023). The field has also begun to build evaluation infrastructure: faithfulness metrics for circuits (Miller et al., 2024), the Mechanistic Interpretability Benchmark for comparing localization methods across tasks and models (Mueller et al., 2025), SAEBench for auditing sparse autoencoder quality (Karvonen et al., 2025), and causal scrubbing for testing mechanistic hypotheses via behavior-preserving resampling (Chan et al., 2022).

These contributions evaluate important objects—methods, circuits, features, alignments, benchmark scores. They do not by themselves answer a different question: what has been established by a mechanistic interpretability *claim*? A claim such as "this head is a name mover," "this feature represents deception," or "this subspace implements a causal variable" is not just a circuit or a metric value. It links a theoretical construct to a measurement procedure, a causal intervention, a scope of generalization, and a natural-language interpretation. Shared metrics are not yet shared validity criteria.

A mechanistic narrative that was retrofitted to match ablation results is currently indistinguishable from one that has survived refutation attempts. This matters because a mechanistic explanation has many adjustable parts—which components to include, what roles to assign, which edges to draw—and a narrative with enough flexibility will always fit the observations. The MechVal framework introduced below provides the criteria needed to distinguish well-supported claims from plausible-sounding ones, making the gaps in evidence visible rather than leaving them implicit.

Several recent results sharpen this point. Sutter et al. (2025) prove that without structural constraints on alignment maps, causal abstraction becomes vacuous: sufficiently expressive maps can perfectly align randomly initialized language models to arbitrary algorithms, achieving 100% interchange-intervention accuracy on tasks the models cannot solve. The SAEBench audit reveals that two of six standard SAE evaluation metrics fail basic diagnostics for reseed stability, ground-truth correlation, and discriminability (Karvonen et al., 2025). And execution-grounded reproducibility checks via MechEvalAgent (Bai et al., 2026) surface claim-evidence coherence issues that narrative review misses. Each finding bears on a different link in the inference chain: alignment-map complexity, measurement reliability, and reproducibility, respectively. What is missing is not any one fix but a principled way to ask, of any claim, which links have been established and which have not.

We introduce MechVal, a formalized epistemology for validating mechanistic interpretability claims, drawing on validation methodology from causal inference, measurement theory, and the philosophy of science. The framework makes interpretability claims explicit, falsifiable, and comparable, formalizing the discovery, validation, and interpretation of causal mechanisms as separate evidential stages. It defines 27 operational criteria across five validity types—construct, measurement, internal, external, and interpretive—in dependency order, with five verdict tiers marking qualitative transitions in evidential status.

MechVal is claim-centric and method-agnostic: it evaluates circuits, SAE features, probing classifiers, steering vectors, transcoders, and crosscoders with the same criteria. A faithfulness score, an interchange-intervention accuracy, a probe accuracy, or a causal-scrubbing result can all serve as evidence within the framework; the framework specifies what kind of evidence each provides, what kind of claim it can support, and what remains untested. MechVal does not replace existing benchmarks or methods. It provides a common layer above them: claim-level validation built on method-level evaluation.

We describe the framework (Section 4), its theoretical foundations (Section 5), its application to 13 published circuits (Section 6), and its implications for the field (Section 7).

2 Related Work

Three recent efforts assess different layers of the MI evidence stack, and MechVal is designed to complement rather than replace them.

The Mechanistic Interpretability Benchmark (MIB; Mueller et al., 2025) compares circuit-discovery and causal-variable-localization methods on a shared task suite using Circuit Model Distance and Interchange Intervention Accuracy. MIB is method-vs-method: given a task, which discovery algorithm produces the best circuit? MechVal is method-agnostic and claim-centric: given a claim about how a model works, what evidence would be needed to sustain it? The two are stacked rather than competing—MIB scores a method's output on a single dimension, while MechVal assesses how that output is interpreted, generalized, and reported.

SAEBench (Karvonen et al., 2025) evaluates sparse autoencoders along six metric families and audits the metrics themselves. Two of six metrics (TPP, SCR) fail diagnostic tests for reseed stability, ground-truth correlation, and discriminability, motivating the explicit measurement-validity dimension in our framework. MechVal treats SAEBench's audit as evidence *about* the M-dimension criteria (M1 reliability, M5 sensitivity, M6 invariance) rather than as a competing benchmark.

MechEvalAgent (Bai et al., 2026) performs execution-grounded reproducibility checks on published MI papers, finding that 93% fail to reproduce when their code is run and that 80% fail claim-evidence coherence checks. Where MechEvalAgent asks "does this code run and match its prose?," MechVal asks "does this evidence pattern support this kind of claim?"

Finally, the LLM-as-judge paradigm (Zheng et al., 2023) has been applied to open-ended evaluation of MI research. However, unstructured LLM evaluation lacks the operational criteria needed to distinguish genuine validity from surface plausibility—the same limitation that affects human narrative review, as demonstrated by MechEvalAgent's finding that narrative review misses issues that execution-grounded checks catch.

A companion paper automates MechVal via structured LLM extraction against the 27 criteria, and describes protocols (curated experiment designs that bundle metrics around specific validity questions) and synthesis algorithms (aggregation across evidence families to produce quantitative scores). Automation and scoring are treated separately because the framework's correctness is independent of whether verdicts are assigned by hand or by machine.

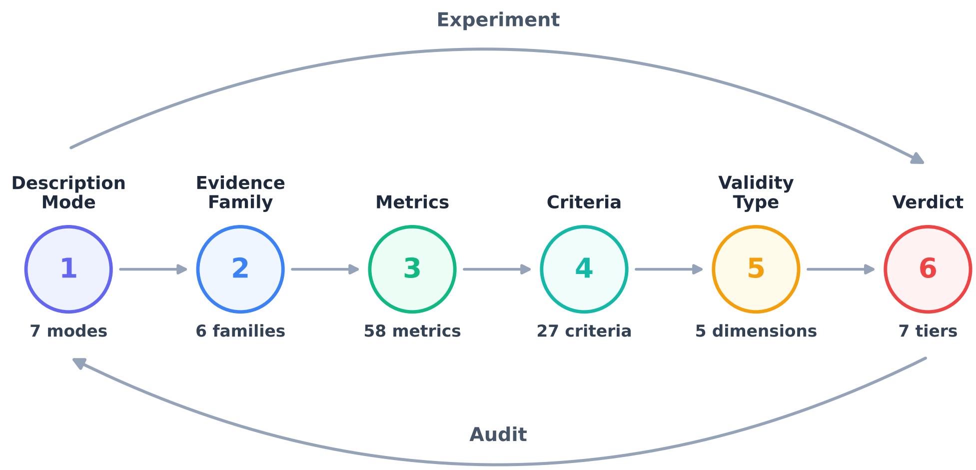

Figure 1: The MechVal framework hierarchy. Six layers from description mode through verdict. **Experiment** (left-to-right): from raw measurements, build upward through criteria and validity types to a verdict. **Audit** (right-to-left): given a claimed verdict, trace backward to identify which lower-layer criteria are unmet.

3 From Claim to Verdict

Consider a concrete claim: "The IOI circuit uses three name-mover heads (9.9, 9.6, 10.0) to copy the indirect object to the output position." This claim appears in a widely-cited paper, is supported by ablation experiments, and has a clear mechanistic narrative. What has this claim established, and what remains untested?

The framework evaluates such a claim through six layers (Figure 1):

1. **Description mode.** At what level is the claim made? The IOI claim operates at the implementational-functional level: it names specific heads and says what each one does. A weaker claim ("GPT-2 solves IOI") would be computational; a stronger one ("head 9.9 implements copying via its OV circuit") would require additional structural evidence.
2. **Evidence family.** What kinds of evidence are available? The original IOI paper provides primarily causal evidence (ablation) and some behavioral evidence (faithfulness). Structural evidence (weight-space analysis of the OV circuits) and cross-model evidence (does the same circuit appear in GPT-2 Medium?) would strengthen the claim by adding independent failure modes.
3. **Metrics.** Which concrete tests were run? Activation patching, role ablation, faithfulness scores. Each metric belongs to an evidence family and produces a measurement that feeds into criteria evaluation.

4. **Criteria.** Does the evidence meet the bar? The IOI circuit passes necessity (I1: ablation degrades performance) and sufficiency (I2: the circuit alone recovers 87% of logit difference). But specificity (I3: does the circuit affect *only* IOI?) is untested, and the 87% faithfulness number is method-conditional—it drops below 50% under non-mean-ablation methods (Miller et al., 2024), partially failing intervention reach (E1).
5. **Validity type.** Which validity dimensions are satisfied? Construct validity (the claim is well-defined), partial internal validity (necessity and sufficiency shown but specificity untested), partial external validity (limited prompt generalization, no cross-model replication). Measurement validity is unaddressed—no calibration of the ablation metrics themselves.
6. **Verdict.** What tier does the claim reach? Mechanistically Supported—necessity and sufficiency are established with at least one method, but the evidence comes primarily from one methodological family (ablation-based), key criteria remain untested (I3, I4, E4), and the faithfulness result is method-conditional. The verdict is not a judgment on the paper's quality; it is a precise characterization of what has been established and what has not.

The 27 criteria also formalize common failure modes observed in published MI research. Table 1 maps twelve such failures to the criteria that detect them.

Table 1: Taxonomy of common failure modes in MI research and the criteria that detect them.

Failure mode	What happens	Criterion	Example
Cherry-picking metrics	Run multiple ablation methods, report only the best number	E1 Intervention reach	IOI: 87% mean ablation reported; resample ablation (40%) omitted
No baseline control	Report high score without testing a random or untrained model	M2 Baseline separation	Nonlinear alignment maps achieve 95% IIA on random models (Sutter et al., 2025)
No negative controls	Only test predictions that should pass; never test what the mechanism should *not* affect	I3 Specificity C1 Falsifiability	IOI: no test of whether the circuit also degrades SVA or factual recall
Construct incoherence	Target construct is not separable from a neighboring one	C4 Discriminant validity	Gender bias circuits: bias and knowledge share the same heads
Off-target effects	Ablation works for the target task; other tasks never checked	I3 Specificity	Knowledge neurons: editing one fact corrupts related facts (ripple effects)
Circuit redefinition	Drop heads that hurt faithfulness, call it "refined"	C1 Falsifiability	Adjusting circuit membership to maximize the target metric
Post-hoc role labeling	Name heads after observed behavior; the label cannot fail	C1 Falsifiability	"Name mover" assigned after seeing logit attribution
No double dissociation	Show necessity but never test the converse	I4 Double dissociation	IOI circuit ablated → IOI breaks; SVA circuit → IOI intact never tested
Interpretive inflation	Label implies intent or algorithm beyond what evidence shows	V4 Anthropomorphism check	"World model" for Othello (evidence supports "linearly decodable")
Scope creep	Validate on one prompt format, claim generality	V5 Scope decl. E2 Prompt gen.	"The IOI circuit" tested only on two-name templates
Flexible scope	Narrow the claim after seeing results so it cannot fail	V5 Scope declaration	Restricting scope post-hoc to the subset of prompts that work
Ignoring alternatives	Circuit works, but so does a different set of heads	I4 Double dissoc. C3 Convergent val.	Meloux et al. find alternative IOI circuits with comparable faithfulness

4 The Framework

The MechVal framework has six layers, presented here in pipeline order (Figure 1): description modes (Section 4.1), evidence families (Section 4.2), metrics (Section 4.3), criteria (Section 4.4), validity types (Section 4.5), and verdict tiers (Section 4.6).

4.1 Description Modes

MI claims operate at different levels of description, following Marr's hierarchy but adapted for transformer circuits. The framework defines seven description modes (Table 2) forming a partial order by evidential commitment. Higher modes require all evidence from lower modes plus bridging evidence that connects levels. A claim at the algorithmic level ("the model runs a detect-inhibit-copy algorithm") requires both the topographic evidence (which heads are involved) and the additional evidence that the heads implement the claimed algorithm. Mode tags are applied to verdicts, not to individual metrics—a single experiment can provide evidence for multiple modes simultaneously.

Table 2: The seven description modes.

	Mode	**What it asks**	**Example**
1	Computational	What function does the system compute, and why?	"GPT-2 predicts the indirect object"
2	Algorithmic	What procedure does it execute?	"A detect-inhibit-copy algorithm"
3	Representational	What information is encoded, and in what geometry?	"The IO name is linearly encoded at layer 8"
4	Impl.-Functional	What transformation does each component perform?	"Head 9.9 copies names to the output"
5	Impl.-Connectomic	How are components wired?	"S-inhibition feeds name movers via residual stream"
6	Impl.-Topographic	Which components are involved?	"Heads 9.9, 9.6, 10.0"
7	Impl.-Statistical	What do activations look like?	"Head 9.9 attends strongly to IO position"

4.2 Evidence Families

An evidence family classifies a metric's output by the kind of signal it produces, independent of the specific tool used to generate it (Table 3). The classification matters because convergent validity—two independent lines of evidence agreeing—is the strongest form of support for a mechanistic claim. But not all agreement is equally informative: two causal metrics agreeing shares failure modes, while a causal ablation and a structural weight-space analysis agreeing is substantially stronger, because a confound that fools one family cannot fool the other.

Table 3: The six evidence families.

	Family	What it measures	Examples
A	Causal	Effects of interventions on computation	Activation patching, DAS-IIA, CATE, mediation
B	Structural	Weight-space properties (no forward pass)	SVD, effective rank, OV/QK composition
C	Information	Information flow and dependencies	Mutual information, PID, OCSE, transfer entropy
D	Behavioral	Input-output behavior under manipulation	Faithfulness, logit diff, KL divergence, dose-response
E	Representational	Geometry of internal representations	CKA, RSA, linear probes, attention entropy
F	Measurement	Trustworthiness of other metrics	Bootstrap stability, seed variance, convergent validity

The framework tracks evidence family coverage explicitly: a claim supported by three families is stronger than one supported by three metrics from the same family.

4.3 Metrics

Metrics are runnable tests that produce measurements about a neural network. They form the empirical base of the framework: evidence families organize them, criteria evaluate them, and verdicts aggregate them. The current implementation includes 84 metrics across families A–E and 14 calibrations in family F. Table 4 illustrates the causal family; the full inventory for all six families is in Appendix A.

Table 4: Causal metrics (family A): effects of interventions on model computation.

Metric	Description
A01 SCM / Pearl	Structural causal models and do-calculus as the formal language for circuit claims
A02 Counterfactual DAS	Distributed Alignment Search and IIA for testing causal abstraction hypotheses
A03 Rubin CATE	Conditional average treatment effects across input subpopulations
A04 Woodward	Interventionist theory of causation applied to circuit ablation
A05 MDC / Glennan	New mechanical philosophy: components organized to produce phenomena
A06 Mediation	Natural direct and indirect effects decomposition for circuit paths
A07 Granger / TE	Observational causal discovery via conditional MI and temporal precedence
A08 PID	Decomposing MI into redundant, unique, and synergistic contributions
A09 MDL / SLT	Minimum description length and singular learning theory for circuit complexity
A10 Regularity / INUS	INUS conditions: insufficient but necessary parts of sufficient sets
A11 Actual Cause	Halpern-Pearl actual causation on specific inputs, not just potential causes
A12 Transportability	Formal conditions for circuit generalization across models and distributions
A13 Causal Discovery	NOTEARS / PC algorithms for learning DAGs over circuit components

4.4 Criteria

Each validity type is operationalized through specific criteria, totaling 27 across the five validity types. Table 5 lists all criteria. Each criterion has a concrete test: a claim either passes, partially passes, or fails each criterion, producing a structured validity profile rather than a single score.

Table 5: The 27 criteria across five validity types.

ID	Criterion	Description
C1	Falsifiability	Can the claim be refuted?
C2	Structural plausibility	Is the mechanism physically possible in the architecture?
C3	Convergent validity	Do multiple independent methods agree?
C4	Discriminant validity	Does the measure distinguish this from neighboring constructs?
C5	Nomological validity	Does the claim fit into a broader theory?
M1	Reliability	Do repeated measurements give the same answer?
M2	Baseline separation	Is the score distinguishable from random/untrained baselines?
M3	Stability	Is the classification robust to perturbation?
M4	Calibration	Are the numbers meaningful?
M5	Sensitivity	Can the instrument detect known-true effects?
M6	Invariance	Does the metric behave consistently across conditions?
I1	Necessity	Is the circuit required for the behavior?
I2	Sufficiency	Is the circuit enough to produce the behavior?
I3	Specificity	Does the circuit do this task and not everything?
I4	Double dissociation	Can this circuit be separated from other circuits?
I5	Confound control	Are alternative explanations ruled out?
E1	Intervention reach	Do different intervention methods agree?
E2	Prompt generalization	Does it work on diverse prompts?
E3	Cross-task generalization	Does the mechanism transfer to related tasks?
E4	Cross-model generalization	Does the mechanism appear in other models?
E5	Graded response	Does partial ablation produce partial effects?
E6	Novel prediction	Does the mechanism predict new, untested behaviors?
V1	Level declaration	At what description mode is the claim made?
V2	Level-evidence match	Does the evidence support claims at that level?
V3	Alternative level	Could the evidence be explained at a different level?
V4	Anthropomorphism check	Is the interpretation projecting human concepts?
V5	Scope declaration	What does the claim explicitly not cover?

4.5 Validity Types

The five validity types form a dependency chain: later types cannot be established without first satisfying earlier ones.

$$\text{Construct} \rightarrow \text{Measurement} \rightarrow \text{Internal} \rightarrow \text{External} \rightarrow \text{Interpretive} \tag{1}$$

Each type addresses a distinct question about the claim:

Construct validity. Is the thing being measured well-defined? Before any measurement is taken, the construct must be specified precisely enough that the claim is falsifiable, structurally plausible, and distinguishable from neighboring constructs. A “deception feature” that cannot be distinguished from an “uncertainty feature” fails construct validity regardless of how carefully it is measured. This draws on philosophy of science (Popper’s falsifiability, Hempel’s operationalism) and psychometrics (construct validity, convergent and discriminant validity).

Measurement validity. Are the instruments trustworthy? Once the construct is defined, the measurement tools used to detect it must be reliable (stable across repetitions), calibrated (separable from random baselines),

Table 6: Verdict tiers with requirements. Each tier subsumes the requirements of all lower tiers.

Tier	Meaning	Requirements beyond previous tier
Proposed	Structural or representational evidence only	Construct validity criteria met (C1–C5)
Causally Suggestive	Necessity shown, sufficiency not established	Internal validity I1 (necessity via ablation)
Mechanistically Supported	Necessity + sufficiency with consistent methods	I1 + I2 (sufficiency), intervention reach (E1), at least 2 ablation variants
Triangulated	Multiple converging lines of independent evidence	Full internal + external + construct criteria from ≥ 3 evidence families
Validated	All five validity types addressed	Measurement + interpretive validity with explicit baselines
Underdetermined	Evidence consistent with multiple mechanisms	Cannot resolve between rival specifications
Disconfirmed	Fails decisively on a key criterion	Negative result

and invariant (consistent across conditions). A metric that gives different answers when run with different random seeds is not evidence. This draws on psychometrics (test-retest reliability, measurement invariance) and pharmacology (assay validation).

Internal validity. Does the evidence support the causal claim? Given a well-defined construct and trustworthy instruments, the evidence must establish that the identified component is causally involved in the behavior—not merely correlated with it. This requires demonstrating necessity (the component is required), sufficiency (the component is enough), and specificity (the component does this task and not everything). This draws on neuroscience (lesion studies, optogenetics, double dissociation) and causal inference (Pearl, 2009).

External validity. Does the mechanism generalize? Causal evidence on one prompt set, with one ablation method, in one model, establishes internal validity at best. External validity requires that the mechanism generalizes across prompts, ablation methods, related tasks, and ideally across models. This draws on pharmacology (Phase III generalization, dose-response) and causal inference (transportability; Pearl and Bareinboim, 2011).

Interpretive validity. Is the interpretation of the mechanism correct? Finally, the natural-language description of the mechanism must be justified by the evidence. A head called a "name mover" based on direct logit attribution carries theoretical implications beyond what the evidence strictly supports. Interpretive validity requires declaring the description level, matching evidence to that level, and checking for anthropomorphic inflation. This is specific to MI but draws on Marr's levels of analysis (Marr, 1982).

4.6 Verdict Tiers

Claims are assigned to one of five verdict tiers representing qualitative transitions in evidential status. Two additional labels handle special cases. The tiers form a strict hierarchy: each requires all evidence from the tier below plus additional evidence.

Every tier carries a specific epistemic meaning: a claim at *Proposed* is not a failed claim but one whose evidential status—and the path to strengthening it—is precisely characterized. The verdict tells researchers exactly what additional evidence would advance the claim: a Proposed claim needs causal intervention (I1) to reach Causally Suggestive; a Causally Suggestive claim needs sufficiency evidence (I2) and method consistency (E1) to reach Mechanistically Supported.

5 Theoretical Foundations

Each component of the framework is grounded in an established validation tradition. The 27 criteria, the dependency chain among validity types, and the verdict tier system are not novel inventions—they are adaptations of principles that have been tested and refined across decades of methodological work in five fields.

5.1 Philosophy of Science

The construct validity criteria (C1–C5) draw on Popper's falsifiability criterion, Mayo's severe testing (Mayo, 2018), and Glennan's new mechanical philosophy (Glennan, 2017). C1 requires that a mechanistic claim state what evidence would disprove it—a requirement routinely violated in MI, where circuit labels are often applied post-hoc without pre-registered predictions. C5 (nomological validity) requires that the claim fit into a broader theoretical network, following the nomological network concept from Cronbach and Meehl.

The verdict tier system is modeled on the hierarchy of evidential support in philosophy of science: observation (Proposed), single-method causal evidence (Causally Suggestive), multi-method causal evidence (Mechanistically Supported), convergent evidence from independent methods (Triangulated), and comprehensive evaluation (Validated).

5.2 Neuroscience

The internal validity criteria (I1–I5) are adapted from neuroscience's toolkit for establishing that a neural circuit implements a computation. I1 (necessity) corresponds to lesion studies: removing the circuit disrupts the behavior. I2 (sufficiency) corresponds to stimulation: activating the circuit alone produces the behavior. I4 (double dissociation) is the gold standard in neuropsychology for establishing functional specificity—showing that circuit A is necessary for task X but not task Y, while circuit B is necessary for task Y but not task X.

The critical distinction between necessity and sufficiency—well-established in neuroscience but often elided in MI—is central to the framework. Most published circuits demonstrate necessity (ablation degrades behavior) but not sufficiency (the circuit alone can produce the behavior). The framework treats this distinction as a qualitative transition between verdict tiers.

5.3 Psychometrics

The measurement validity criteria (M1–M6) draw on psychometric principles of test construction. M1 (reliability) is test-retest reliability: the same metric applied to the same model should give the same answer. M6 (invariance) is measurement invariance: the metric should behave consistently across conditions (e.g., different prompt sets, different ablation methods). The calibrations (F01–F15 in the implementation) operationalize these principles as concrete statistical tests.

The distinction between convergent and discriminant validity (C3 and C4) comes directly from Campbell and Fiske's multitrait-multimethod matrix. A valid construct should show convergent validity (multiple methods agree) and discriminant validity (the construct is distinguishable from related constructs). In MI terms: a "deception feature" should be detected by both probing and activation patching (convergent), and should not also fire on uncertainty, hedging, or politeness (discriminant).

5.4 Pharmacology

The external validity criteria, particularly E5 (graded response), draw on pharmacology's dose-response methodology. A causally relevant component should produce graded effects: partial ablation should produce partial behavioral change, and the relationship should be monotonic or at least systematic. The framework treats dose-response as a minimum bar for causal claims about magnitude—a requirement absent from most MI evaluations, which typically report binary ablation (all-or-nothing).

The three-phase protocol used in the case studies (measurement reliability, causal validity, generalization) mirrors the phase structure of clinical trials: Phase I establishes safety and dosing (analogous to measurement calibration), Phase II establishes efficacy (causal validity), and Phase III establishes generalization.

5.5 Causal Inference

The internal validity criteria (I1–I5) draw on causal inference methodology (Pearl, 2009; Spirtes et al., 2000). The necessity/sufficiency distinction maps to interventionist causation (Woodward, 2003): necessity (I1) corresponds to "would the behavior persist if the component were absent?" and sufficiency (I2) to "would the component alone produce the behavior?" The external validity criteria, particularly E4 (cross-model generalization), draw on Pearl and Bareinboim's transportability theory (Pearl and Bareinboim, 2011)—formal conditions under which causal findings transfer across settings.

6 Case Studies

We apply the framework to 13 published MI results, evaluating each through all five validity lenses. The case studies are not intended to rank papers—they demonstrate what the framework looks like in practice and show that the tier system discriminates between well-validated and poorly-validated claims. Quantitative per-paper scoring—including the Claim Validity Score, tier-agreement confusion matrix, and per-criterion heatmaps—is the subject of a companion paper; here we keep the analysis qualitative and trace each verdict to specific missing or present evidence.

Figure 2: Per-dimension validity profiles for ten selected circuits spanning four verdict tiers: Triangulated (8.0–10.0), Mechanistically Supported (6.0–7.9), Causally Suggestive (4.0–5.9), and Proposed (<4.0). Each radar shows five validity dimensions; the progressive collapse from full pentagons to collapsed shapes makes the tier boundaries visually concrete.

6.1 Verdict Assignments

Table 7 presents the verdict assignments. The key finding is that the tier system produces meaningful discrimination: the 13 claims span the full range from Proposed to Triangulated, with verdict differences traceable to specific missing evidence rather than arbitrary scoring.

Table 7: Verdict assignments for 13 published circuits. Verdicts are based on published evidence as of evaluation date.

Verdict	Circuit	Key limiting criterion
Triangulated	Induction Heads	Simple mechanism, broad replication, thick nomological network
Mechanistically Supported	IOI Circuit	Method-conditional faithfulness (87% mean ablation, <50% other methods); specificity untested
	Greater-Than	Strong structural plausibility (C2); limited prompt generalization
	Copy Suppression	Unusually clean specificity; sufficiency partially established
Causally Suggestive	Grokking	Validated within toy scope; toy-to-real gap is the limitation
	Successor Heads	Cross-domain generalization as convergent evidence; ablation method-conditional
	Docstring Circuit	Label ambiguity: "variable binding" vs. simpler "positional copying"
	SAE Features	Thin nomological network; features may be dictionary properties, not model properties
	Othello World Model	"World model" label exceeds evidence ("linearly decodable"); interpretive inflation
Proposed	Knowledge Neurons	Strong intervention but weak mechanistic story
	Superposition	Validated theory in toy models; awaits real-model confirmation
	Probing Classifiers	Decodability without intervention = no internal validity
	Gender Bias Circuits	Construct incoherence: bias and knowledge share circuits

6.2 Illustrative Examples

We describe three case studies in detail to illustrate how the framework produces its verdicts.

Induction Heads (Triangulated). Olsson et al. (2022) identify a two-head composition mechanism for in-context copying: previous-token heads attend to the token before a repeated sequence, and induction heads use this signal to predict the next token. The claim reaches Triangulated because it satisfies criteria across multiple independent evidence families. The mechanism has been replicated across model scales and architectures (E4), confirmed through both activation-level and weight-level analysis (C3 convergent validity from different families), and produces clean dose-response effects (E5). The nomological network is thick: the mechanism connects to in-context learning theory, training dynamics, and cross-model structural predictions. The relative simplicity of the mechanism (two functional roles, clear compositional structure) makes each criterion easier to satisfy.

IOI Circuit (Mechanistically Supported). Wang et al. (2023) identify 26 attention heads forming a six-role circuit for indirect object identification. The circuit demonstrates strong necessity (I1) and sufficiency (I2): running the model with everything outside the circuit mean-ablated recovers 87% of the full model's logit difference. However, Miller et al. (2024) show that this 87% figure is specific to mean ablation; under other ablation methods, faithfulness drops below 50%. This method-conditional result partially satisfies E1 (intervention reach) but reveals that the causal claim is weaker than the headline number suggests. Specificity (I3) is untested: the authors do not report whether ablating the IOI circuit degrades unrelated tasks. A formal double-dissociation has not been conducted. The circuit reaches Mechanistically Supported rather than Triangulated because the evidence comes primarily from one methodological family (ablation-based) and key criteria (I3, I4, I5, E4) remain unaddressed.

Probing Classifiers (Proposed). Linear probing (Belinkov, 2022) trains a classifier on activations to test whether a concept is linearly decodable. The core limitation is fundamental: *a measurement without an intervention is a measurement without internal validity.* Probe success establishes that information is decodable from the activation space, but not that the model uses that information during inference. High probe accuracy is consistent with genuine encoding, incidental linear separability in high-dimensional space, and confound encoding (a correlated feature rather than the target). Without causal follow-up (DAS, intervention along the probe direction), the claim cannot advance beyond Proposed. Probing *with* causal follow-up can reach Causally Suggestive; with additional control tasks and cross-method convergence, it can reach Mechanistically Supported. The framework does not dismiss probing—it precisely characterizes what probing alone does and does not establish.

6.3 Patterns Across Case Studies

Several patterns emerge from evaluating these claims side by side.

Sufficiency gap. Most circuits demonstrate necessity (I1) but not sufficiency (I2). The I1→I2 transition—from "removing this breaks the behavior" to "this alone produces the behavior"—is the most common barrier between Causally Suggestive and Mechanistically Supported.

Method-conditional results. The IOI circuit's headline numbers are specific to mean ablation. Ablation type is part of the causal claim, not an implementation detail. The framework captures this through E1 (intervention reach), which requires agreement across ablation methods.

Toy-model ceiling. Grokking and Superposition achieve high validity within toy scope—but the gap between toy-model proof-of-concept and real-model confirmation remains the field's central challenge. The framework handles this through the scope declaration criterion (V5): a claim Validated within a declared scope is a legitimate result, not a failure.

Interpretive inflation. Labels like "world model," "knowledge neuron," and "deception feature" carry theoretical implications beyond what the evidence supports. The framework systematically identifies where labels exceed evidence through V4 (anthropomorphism check) and V5 (scope declaration).

Construct incoherence. Gender Bias Circuits fail not because evidence is lacking but because the construct itself—"bias" separable from "legitimate gender processing"—may not be well-posed. Sometimes the right answer is "this question is not well-defined," not "we need more data." The framework captures this through the Construct validity criteria, which must be satisfied before any measurement is meaningful.

7 Discussion

7.1 Method Agnosticism

The framework evaluates any MI claim type using the same criteria. It applies to:

- **Circuits:** Sets of attention heads, MLPs, and edges implementing a computation.
- **SAE features:** Directions learned by sparse autoencoders.
- **Probing classifiers:** Linear classifiers trained on activations.
- **Steering vectors:** Directions used for activation addition/subtraction.
- **Transcoders:** Cross-layer feature extractors.
- **Crosscoders:** Multi-model feature extractors.

This agnosticism is not a design choice—it is a consequence of grounding the framework in validity theory rather than in the specifics of any detection method. The questions "Is the construct well-defined?" and "Is the evidence causal?" apply regardless of whether the evidence comes from ablation patching, dictionary learning, or linear probing.

7.2 Relationship to Existing Benchmarks

The Mechanistic Interpretability Benchmark (MIB; Mueller et al., 2025) provides competitive benchmarks for circuit discovery. MechVal is complementary: MIB evaluates how well a method discovers circuits; MechVal evaluates how well-validated the discovered circuit is. A circuit that achieves high CMD on MIB may still be poorly validated if it lacks sufficiency evidence, dose-response testing, or cross-method consistency. The two frameworks can be composed: discover a circuit via MIB-style methods, then evaluate it via MechVal's validity criteria.

SAEBench evaluates SAE quality via standardized metrics. MechVal's measurement validity criteria (M1–M6) and calibrations (F01–F15) address a prior question: are the SAEBench metrics themselves trustworthy? The SAEBench audit's finding that TPP and SCR fail basic diagnostics is precisely the kind of measurement validity failure that the framework's calibrations are designed to catch.

7.3 Limitations

Scope. The current implementation is applied to GPT-2 Small. Extension to larger models requires substantially more compute and may reveal that some criteria (e.g., E4 cross-model generalization) are harder to satisfy than the small-model case studies suggest.

Verdict assignment subjectivity. While the 27 criteria are operationally defined, the aggregation from criterion-level verdicts (pass/partial/fail) to tier-level verdicts involves judgment. Two evaluators could assign different tiers to the same circuit if they weight criteria differently. The framework mitigates this by requiring criterion-level transparency: the structured validity profile is the primary output, and the tier is a summary.

No novel experiments. This work is a theoretical contribution. The case study verdicts are based on published evidence, not new experiments. Running the framework's full metric suite on all evaluated circuits with actual model interventions is future work that requires GPU compute.

7.4 Implications for AI Safety

The validity gaps described in Section 1 have direct implications for AI safety. Organizations are beginning to deploy internal monitors based on MI findings—probes that flag potential deception, steering vectors that suppress harmful outputs, circuit-level classifiers that trigger human review. If those monitors are built on unvalidated features, they provide false confidence: the system appears monitored when it is not.

The framework provides a concrete protocol for assessing monitor trustworthiness. A "deception feature" can be evaluated through the same criteria as any other MI claim: Is it well-defined (C1–C5)? Is it reliably measurable (M1–M6)? Is it causally necessary and sufficient (I1–I2)? Does it generalize across prompts and models (E1–E6)? Does the "deception" label match the evidence (V1–V5)?

The verdict tier system provides deployment guidance. A feature at Proposed should not be deployed as a monitor. A feature at Causally Suggestive provides preliminary evidence but lacks sufficiency testing. A feature at Mechanistically Supported or above has the minimum evidential basis for cautious deployment with ongoing validation. This is not a guarantee—it is a calibrated assessment of evidential status, which is what deployment decisions require.

8 Conclusion

The mechanistic interpretability field has made remarkable progress in identifying circuits, features, and representations in neural networks, and has begun to build serious evaluation infrastructure for comparing methods. But shared metrics are not yet shared validity criteria: the field evaluates methods and artifacts without a common protocol for evaluating the claims built on them.

The MechVal framework addresses this gap. Its contribution is not a new detection method but a systematic protocol for determining what kind of evidence supports a given claim and what remains untested. The 27 criteria, five validity types, five evidence families, and five verdict tiers provide a common language for evaluating MI claims—whether they concern circuits, SAE features, probes, or steering vectors.

The framework's application to 13 published circuits demonstrates that the tier system discriminates meaningfully. The discrimination is informative: verdict differences are traceable to specific missing evidence, and each verdict tells researchers exactly what additional evidence would advance the claim. The framework does not rank papers or methods. It characterizes the evidential status of claims, making the gaps visible and the path forward explicit.

The framework is open-source and method-agnostic. Its contribution is a public good: the measurement infrastructure that makes interpretability claims trustworthy.

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A Metrics by Evidence Family

Table 8: Causal metrics (A): effects of interventions on model computation.

Metric	**Description**
A01 SCM / Pearl	Structural causal models and do-calculus as the formal language for circuit claims
A02 Counterfactual DAS	Distributed Alignment Search and IIA for testing causal abstraction hypotheses
A03 Rubin CATE	Conditional average treatment effects across input subpopulations
A04 Woodward	Interventionist theory of causation applied to circuit ablation
A05 MDC / Glennan	New mechanical philosophy — components organized to produce phenomena
A06 Mediation	Natural direct and indirect effects decomposition for circuit paths
A07 Granger / TE	Observational causal discovery via conditional MI and temporal precedence
A08 PID	Decomposing MI into redundant, unique, and synergistic contributions
A09 MDL / SLT	Minimum description length and singular learning theory for circuit complexity
A10 Regularity / INUS	INUS conditions — insufficient but necessary parts of sufficient sets
A11 Actual Cause	Halpern-Pearl actual causation on specific inputs, not just potential causes
A12 Transportability	Formal conditions for circuit generalization across models and distributions
A13 Causal Discovery	NOTEARS / PC algorithms for learning DAGs over circuit components

Table 9: Structural metrics (B): weight-space properties requiring no forward pass.

Metric	Description
B01 SVD / Spectral	Singular value decomposition to identify dominant computational directions
B02 Effective Rank	Entropy-based dimensionality as a scalar summary of spectral concentration
B03 OV / QK Decomp.	Decomposing attention heads into OV (what to write) and QK (where to attend)
B04 Weight Alignment	Cosine similarity between principal weight directions across heads
B05 Norm Trajectory	Spectral norm ratios tracking signal amplification through components
B06 Template Distance	Graph-edit and metric distances between circuits discovered for different tasks
B07 Polysemanticity	Measuring whether components encode multiple unrelated features in superposition
B08 ICA / NMF	Independent component analysis for decomposing weights into interpretable parts
B09 Weight Classifier	Training classifiers on weight matrices to predict circuit membership

Table 10: Information-theoretic metrics (C): information flow and dependencies.

Metric	Description
C01 Mutual Info.	Total shared information between circuit components and task performance
C02 Conditional MI	Information shared after conditioning on other parts of the circuit
C03 Transfer Entropy	Directed information flow between components across layers
C04 PID	Decomposing shared information into unique, redundant, and synergistic atoms
C05 Info. Bottleneck	How efficiently circuits compress input while preserving task-relevant signal
C06 O-Information	Whether a group of components interacts redundantly or synergistically
C07 Granger Causality	Whether one component's past activations improve prediction of another's future
C08 OCSE	Estimating causal influence between components using only observational data
C09 NOTEARS	Learning directed acyclic graph structure from observational activation data

Table 11: Behavioral metrics (D): input-output behavior under manipulation.

Metric	**Description**
D01 Faithfulness	Whether an identified circuit faithfully reproduces the full model's behavior
D02 Logit Diff	How much of the model's logit difference the circuit recovers
D03 KL Divergence	Information-theoretic distance between circuit and full model distributions
D04 CE Delta	Change in cross-entropy loss when the circuit is ablated
D05 Top-K Accuracy	Whether the circuit preserves the model's top-K predicted tokens
D06 Cross-Task	Whether a circuit discovered on one task transfers to a different task
D07 Cross-Scale	Whether circuit structure replicates in larger or smaller models
D08 Prompt Paraphrase	Circuit consistency across semantically equivalent prompt templates
D09 Generalization Gap	Sensitivity of circuit discovery to hyperparameters and methodological choices

Table 12: Representational metrics (E): geometry and content of internal representations.

Metric	**Description**
E01 DAS-IIA	Whether a learned linear subspace causally encodes a target variable
E02 Linear Probe	Whether a target variable is linearly decodable from intermediate representations
E03 RSA	Comparing representation geometry by correlating pairwise distance matrices
E04 CKA	Kernel alignment for cross-layer and cross-model representation comparison
E05 Subspace Align.	Cosine alignment between SVD-derived principal directions of weight matrices
E06 PCA Dim.	Effective dimensionality of circuit subspaces via activation covariance spectrum
E07 Intrinsic Dim.	True manifold dimensionality, connecting to geometric complexity
E08 Participation Ratio	How many dimensions are effectively active in a representation
E09 Persistent Homology	Topological data analysis detecting loops and voids in activation manifolds
E10 Cross-Task Overlap	Representational structure shared between tasks via IIA transfer

B Example Circuit Specification

The IOI claim specification defines a 6-step computational DAG:

1. **Duplicate Token Detection** (DTH: heads 0.1, 3.0) — detects that a name appears twice.
2. **Previous Token Tracking** (PTH: heads 2.2, 4.11) — attends to the token before a name.
3. **Induction** (IND: heads 5.5, 6.9) — uses DTH and PTH signals to identify the repeated name.
4. **S-Inhibition** (S-Inh: heads 7.3, 7.9, 8.6, 8.10) — suppresses the repeated subject name.
5. **Name Mover** (NM: heads 9.9, 9.6, 10.0) — copies the non-repeated (indirect object) name to the output.
6. **Negative Name Mover** (NegNM: heads 10.7, 11.10) — provides opposing signal.

Positive predictions (7).

- Ablating DTH reduces output (≥ 0.2 decrease in logit difference).
- Ablating induction heads substantially reduces logit difference (≥ 0.5).

- Ablating S-Inh kills output (≥ 0.8 decrease).
- Ablating S-Inh reduces name mover activation (≥ 0.2).
- Ablating NegNM increases logit difference (removes opposing signal).
- Ablating DTH reduces induction head activation (tests DTH$\rightarrow$IND edge).
- Ablating PTH reduces induction head activation (tests PTH$\rightarrow$IND edge).

Negative controls (5).

- Ablating NM does not affect upstream S-Inh.
- Ablating NegNM does not affect upstream S-Inh.
- Ablating PTH alone does not destroy circuit output (< 0.3 decrease).
- Ablating S-Inh does not affect upstream DTH.
- Ablating NM does not affect NegNM (parallel roles, not connected).

This specification encodes the directed causal structure of the proposed mechanism and provides concrete, falsifiable predictions that can be tested automatically by the verification pipeline.